

ANTON MELNYCHUK

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EDUCATION

Yale University

B.S. in Electrical Engineering & Computer Science
Ph.D. in Computer Science (Sterling Prize Fellow)

New Haven, CT

May 2026

Deferred

SUMMARY

Passionate about hardware-software co-design for *data-intensive* workloads. Hands-on experience spanning RTL design and verification, RTL-to-GDSII physical design, async and multi-clock interconnects, FPGA prototyping, and bring-up.

Languages: SystemVerilog/Verilog, VHDL, C/C++, CUDA, Rust, Python, Tcl, Perl.

RTL/DV: AXI4, NoC/protocol bridges, CDC & async interfaces, SVA/UVM, formal.

PD: Cadence Genus, STA/timing closure, Magic VLSI, Virtuoso, sign-off.

Systems: PCIe/DMA, HBM/LPDDR6, SerDes/UCIe, AMD Zynq/Versal, Vivado, Nsight, Linux.

WORK EXPERIENCE

Stealth Startup

Member of Technical Staff

San Jose, CA

Sep. 2026 – Present

Microsoft Research Asia

Research Collaborator (External)

Remote

Mar. 2026 – Present

- Investigated memory architectures for LLM inference, where model and KV-cache sizes exceed HBM capacity.
- Exploring approaches to alleviate the memory-bandwidth bottleneck without sacrificing memory capacity.
- Nsight-profiled Qwen3-235B memory-access graphs, sized the local working set, designed RTL for the Tensor Addressable Bridge (TAB) prefetch unit (at [FengHuang](#)) with 50% GPU reduction at equal end-user latency.
- Contributed to vendor-feasibility discussions, UCIe/SerDes lane budgets, and packaging constraints.

Riverlane Ltd.

Quantum Research Intern

Greater Boston, MA

Summer 2026

- Optimized CUDA persistent kernels and LER-tested two GPU-based QEC decoders (surface and qLDPC codes) under phenomenological and circuit-level noise; identified the dominant sources of decode latency.
- FPGA-ported NVIDIA Ising AI pre-decoder with a sub-1 μ s hardware-accelerated global matching solver over PCIe.
- Presented stage-synchronous, memory-local *programmable* QEC accelerators (CGRAs, SambaNova) that meet throughput and latency guarantees under local to non-local syndrome connectivity (IEEE Quantum Week poster).

Asynchronous VLSI Lab

Undergraduate Research

New Haven, CT

Jul. 2025 – Aug. 2026

- RTL-to-GDSII design of [HALO](#), an iEEG closed-loop seizure-neuromodulation brain-computer interface.
- Built the post-silicon validation platform: PCB design support, power delivery, Intan-based neural data acquisition, electrode interfaces on FPGA, and software visualization tools; in-vivo tested at Yale School of Medicine.
- Designed and verified the asynchronous NoC: two RTL tiles (detection and stimulation) linked by bundled-data handshake channels, with an AXI4 bridge DMA streaming iEEG samples and flags over Ethernet.
- Reduced false-positive seizure detections versus the NeuroPace® baseline; treatment-efficacy studies ongoing.

Yale University

Teaching Assistant, CPSC 3230 Systems Programming & Computer Organization

Spring 2025 – Fall 2026

Teaching Assistant, CPSC 4200/ECE 4201 Computer Architecture

Spring 2026

- Supported the PARCv2 pipelined-processor labs, debugging students' SystemVerilog designs and RTL sim flows.
- Designed and built a Rust-based teaching OS, adopted as the core project for 200+ students annually at Yale.

SELECTED PROJECTS

Longhorn Silicon, UT Austin AI Inference Chip Design

May 2026 – Present

- Contributed RTL to the KV Cache Engine's compress-on-write / decompress-on-read datapath between the attention unit and LPDDR5X, cutting off-chip KV bandwidth $\sim 3.8\times$ near-losslessly.
- KVE+TIU coprocessor hardened DRC/LVS-clean on GF180 (0.262 mm², SPI host interface).

MOS 6502 CPU, Full-Custom Physical Design

Oct. 2025 – Dec. 2025

- Full-custom layout of the [MOS 6502 8-bit CPU](#) (Apple II, NES) in Magic VLSI on a 90 nm SCMOS flow.
- Designed a KiCad test board with VGA output and a custom 6T SRAM (~ 512 KB); hand-assembled and validated.
- Coordinating silicon test of IHP SG13G2 OpenPDK 6502 tapeouts with W. Grabinski (IHP/MOS-AK).